Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/05/2025
PI Name:	Flickinger, Katharyn Ph.D.		
Project Title:	Metabolic Measurement, Manipulation, and Countermeasure Strategies		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Food and Nutrition**:Risk of Performance Decrement and Crew Illness Due to Inadequate Food and Nutrition		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 TRISHRFA2401PD: Translational Research Institute for Space Health (TRISH) Announcement Soliciting Postdoctoral Fellowship Applications
Start Date:	09/01/2024	**End Date:**	11/30/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2026 per TRISH (Ed., 4/3/26).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Callaway, Clifton Ph.D. (MENTOR: University of Pittsburgh)		
Grant/Contract No.:	NNX16AO69A-P0902		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Upcoming Artemis missions will be sending astronauts back to the lunar surface as a launching point for future exploration of Mars. Long-duration spaceflight exposes astronauts to a multitude of hazards that put both the health of crewmembers, and the mission's success, at risk. Decades of short-duration space missions and missions in low Earth orbit have demonstrated that the taxing environment in space affects multiple physiological systems. The musculoskeletal system has notable declines due to the lack of mechanical loading supplied by Earth's gravity. Weight-bearing muscles atrophy rapidly and shift towards fast-twitch muscle fibers, thus reducing their strength and endurance. This decline in muscle health reduces astronaut readiness both during spaceflight and upon returning to Earth. Outside of space travel, muscle loss due to muscle unloading also afflicts persons with serious medical illness, for whom deconditioning after forced bedrest is a major barrier to recovery and rehabilitation. In-flight countermeasures can mitigate the decline in muscle health, yet some decline still occurs, with the risk currently rated as "Yellow: accepted with optimization" within the NASA Human Research Program (HRP) Human Research Roadmap. In upcoming Artemis missions to the lunar space and surface and in future long-duration missions, astronauts will have less time and physical space dedicated to exercise than on the International Space Station (ISS). Thus, we must test whether we can optimize muscle health with limited exercise equipment and shorter durations of exercise time immediately for the Artemis missions and in preparation for ever-increasing mission lengths as we progress towards Mars. My independent research program will study the impact of global metabolic rate on musculoskeletal health, with the space-related goal of augmenting existing countermeasures to muscle loss in microgravity. I will conduct a series of studies in healthy persons during bedrest		

	and in hospitalized patients with serious illness to examine the relationship between global metabolic rate and musculoskeletal health. To maintain the utility of our measures for spaceflight and to be useful in many healthcare settings, we will validate portable and non-invasive tools to measure metabolic demand and muscle health. Aim 1: Refine non-invasive physiologic measurements for estimating metabolic rate and muscle health in both healthy and patient populations. Aim 2: Determine how spontaneous variation and pharmacological lowering of global metabolic rate relates to musculoskeletal health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To maintain the utility of our measures for spaceflight and to be useful in many healthcare settings, we will validate portable and non-invasive tools to measure metabolic demand and muscle health.
Task Progress:	Within this first year, we have been successfully collecting extensive data on metabolic rate in healthy individuals in both short-term and long-term studies with and without pharmacologically induced reduction in metabolism. Throughout these protocols, we have been using a variety of small-profile devices to allow for non-tethered, remote monitoring of participants, as well as collecting spot-indirect calorimetry to evaluate how accurate estimations from the various devices compare to currently accepted methods of measurement. For the longer 5-day protocols, we have also used doubly-labeled water (DLW) to measure total energy expenditure before and during 5 days of pharmacologically induced metabolic reduction. Devices used include: Actigraphs, LifeWare ProPatch, and BodyCap medical eCelcius core temperature pill. During the 5-day protocol, we learned that drug tolerance is frequently developed after the third night, and participants experienced mild withdrawal symptoms. Analyses from the DLW indicate that there is still metabolic savings despite the drug tolerance. We are now in the process of preparing manuscripts describing the estimated payload savings (food, oxygen consumption, carbon dioxide production, astronaut ins and outs, etc.) that would occur from an induced metabolic reduction. We have been pleased that most participants were able to successfully complete some, if not all the daily exercise portion of the protocol. Preliminary data analyses suggest that there is an immediate change in muscle composition (assessed via electrical impedance myography) of these participants, and there is a unique shift in hunger, thirst, and satiety. We will continue to study these findings in further detail. Our in-hospital protocol quantifying changes in muscle health and movement patterns of post cardiac arrest patients, had several challenges and technical hurdles that delayed enrollment. We were able to overcome the barrier of obtaining consent from a legally authorized representative (medical decision maker), by requesting the ability to obtain virtual, electronic consent. This made a huge improvement in our ability to enroll participants within a timely enrollment window. We have experienced multiple equipment failures with the Myolex electrical impedance device- requiring frequent replacement of parts. We have opted to continue with this device rather than switching brands so as not to confound the study data and are working with the manufacturer to improve our device. We will be looking for alternatives for future studies. In the meantime, we are asking the company to have spare pieces on hand, so that if something breaks, we can repair it ourselves and continue with uninterrupted data collection. During the next year we will: 1) continue our series of shorter studies to observe spontaneous variation in metabolism in our laboratory setting 2) complete enrollment for our hospital-based study and begin data analysis. Additionally, we are planning a series of lab studies to investigate isometric exercise and inspiratory muscle strength training and their impact on both metabolic rate and musculoskeletal health and their use as a countermeasure to the discomforts associated with the transition into and out of microgravity, including fluid shifts.
Bibliography Type:	Description: (Last Updated: 02/19/2026)
Articles in Peer-reviewed Journals	Wu V, DeMaio R, Jonsson A, Monteleone J, Flickinger K, Weissman A, Guyette F, Buysse D, Birabaharan J, Empey P, Callaway C. "Examining the relationship between core temperature, heat balance, and energy expenditure." Ther Hypothermia Temp Manag. 2025 Oct 13. Online ahead of print. https://doi.org/10.1177/21537658251387188 ; PMID: 41078278 , Oct-2025
Articles in Peer-reviewed Journals	Neofes D, Wu V, Monteleone J, Sheng T, Ortega-Reyes O, Salinas M, Hierro M, DeMaio R, Weissman AJ, Guyette FX, Gordon B, Mortreux M, Melanson K, Buysse D, Empey P, Ward-Ritacco C, Callaway CW, Flickinger KL. "Changes in grip strength during 5-day metabolic reduction." Med Sci Sports Exerc. 2025 Oct;57(10S):798-9. https://doi.org/10.1249/01.mss.0001161812.43889.d1 , Oct-2025
Articles in Peer-reviewed Journals	Flickinger KL, Weissman AJ, Guyette FX, DeMaio R, Jonsson A, Wu V, Monteleone JL, Zurowski EA, Birabaharan J, Buysse DJ, Empey PE, Nolin TD, West RE, Callaway CW. "Sustained metabolic reduction and hypothermia in humans." PLoS One. 2025 Apr 21;20(4):e0321117. https://doi.org/10.1371/journal.pone.0321117 ; PMID: 40258026; PMCID: PMC12011254 , Apr-2025
Significant Media Coverage	Doerfler, A. (Flickinger interview) ""To the stars and back: What space biomedicine brings home." Dr. Flickinger and Dr. Calloway interviewed for University of Pittsburgh publication." Pitt Medicine Magazine, https://www.pittmed.pitt.edu/news/space-biomedicine-research-patients-on-earth-afshin-beheshti-radiation-mitochondria-bone-loss-hibernatic , Jul-2024
Significant Media Coverage	Jefferson B. ""Summary of postdoc work." Summary in University of Pittsburgh publication." PittWire Magazine., Dec-2024